

Classes and Objects

What is a Class?

A **class** is a blueprint or prototype that defines the data and functions common to all objects of a certain kind. It is a user-defined data type, also known as an Abstract Data Type (ADT).

A class is a collection of:

- **Data members** — the variables that store data
- **Member functions** — the functions that operate on that data

Think of a class like a cookie cutter — it defines the shape, but it is not the cookie itself.

```
class Student {
private:
    int roll;
    char name[20];
public:
    void getData() {
        cout << "Enter roll number and name: ";
        cin >> roll >> name;
    }
    void putData() {
        cout << "Roll: " << roll << ", Name: " << name;
    }
};
```

What is an Object?

An **object** is an instance of a class — the actual entity created from the blueprint. While a class is just a definition, an object is what gets created in memory when your program runs.

```
Student s;           // s is an object of the Student class
s.getData();         // calling a member function on the object
s.putData();
```

Access Specifiers

Access specifiers control who can access the members of a class. There are three types:

- **private:** Members can only be accessed by functions inside the class. This is the default if no specifier is given. Used for data hiding.
- **public:** Members can be accessed from anywhere — inside or outside the class.
- **protected:** Access lies between private and public. Used primarily in inheritance — derived classes can access protected members of their base class.

```
class BankAccount {
private:
```

```
    double balance;          // hidden from outside
public:
    void deposit(double amount) { balance += amount; } // accessible
    double getBalance() { return balance; }           // accessible
};
```

Memory Allocation for Objects

- Memory is allocated when an object is **declared**, not when the class is defined.
- All objects of a class **share the same member functions** — only one copy of each function exists in memory.
- Each object gets its **own copy of the data members**.

Accessing Members

The **dot operator (.)** is used to access members of an object:

```
objectName.functionName(arguments);
```

The **scope resolution operator (::)** is used to define member functions outside the class body:

```
void Student::getData() {
    cin >> roll >> name;
}
```